Science of the Total Environment xxx (xxxx) xxx-xxx

Contents lists available at ScienceDirect

Science of the Total Environment

journal homepage: www.elsevier.com

Particulate matter mitigation via plants: Understanding complex relationships with leaf traits

Zhong Yu Chiam [a,*], Xiao Ping Song [b,c], Hao Ran Lai [a,d], Hugh Tiang Wah Tan [a]

[a] Department of Biological Sciences, National University of Singapore, 14 Science Drive 4, 117542, Singapore
[b] Department of Architecture, National University of Singapore, 4 Architecture Drive, 117566, Singapore
[c] ETH Zurich, Future Cities Laboratory, Singapore-ETH Centre, 1 Create Way, Create Tower, #06-01, 138602, Singapore
[d] Yale-NUS College, 16 College Avenue West, 138527, Singapore

ARTICLE INFO

Article history:
Received 24 April 2019
Received in revised form 13 June 2019
Accepted 17 June 2019
Available online xxx

Editor: Elena Paoletti

Keywords:
Wind tunnel
Southeast Asia
Surface deposition velocity
Boundary layer
Gravitational sedimentation
Phytoremediation

ABSTRACT

Elevated levels of airborne particulate matter (PM) pose health risks to populations living in many cities worldwide. To remediate the impact of air pollution, urban greening has been increasingly explored as a possible way to remove PM from the surroundings. However, existing research focuses mainly on species-specific assessments within temperate climates that may not necessarily grow outside of their local regions. To address sampling limitations associated with the great species diversity in the tropics, our study quantified three key functional traits—leaf hairiness, average leaf area (ALA) and specific leaf area (SLA)—each spanning a wide range of values across 20 tropical species. Wind-tunnel experiments were used to assess surface deposition velocity on leaves; the size fractions PM_{10} and $PM_{2.5}$ were measured through multiple rounds of filtration and gravimetric analysis. The effect of upper- and lower-surfaces of leaves on deposition velocity and their interactions with hairiness were also investigated. Results show that greater upper-surface hairiness and a low SLA were consistent predictors of higher deposition velocity for both PM-size fractions. Hairs on leaves serve as obstacles while low SLA tends to be associated with smaller and thicker leaves, together favoring the deposition and retention of PM. The possible mechanisms behind important plant traits and their interactions are discussed. By testing quantifiable effects of specific plant traits, we provide generalizable findings that may be applied to urban greening efforts. Future work to consider other gaseous pollutants and plant-scale effects can help ensure a more comprehensive evaluation of plant suitability for pollutant mitigation.

© 2019.

1. Introduction

Air pollution poses a major health risk in many countries around the world, associated with diseases such as stroke, ischemic heart disease, chronic obstructive pulmonary disease, acute lower respiratory disease and lung cancer (WHO, 2016). Among the major air pollutants used as indicators of air quality, particulate matter (PM) was found to be one of the most harmful constituents (Quah and Boon, 2003; WHO, 2006). In 2012, pollution by ambient PM was responsible for 3 million deaths, with 27% occurring within Southeast Asia alone (WHO, 2016). PM is a generic term used to describe a mixture of small particles and liquid droplets ranging from 0.001 to 100 μm in aerodynamic diameter (D_p) (Popek et al., 2015), where size is associated with the degree of penetration and damage caused to the human pulmonary system (Koneig, 2000). PM does not only originate from natural sources such as volcanic eruptions and wildfires, but also anthropogenic emissions from vehicular and industrial sources (Dzierżanowski et al., 2011; Marcazzan et al., 2001). When inhaled, large particles (PM_{100}, D_p = 10–100 μm) are primarily deposited in the periphery of the upper respiratory tract, while coarse particles (PM_{10}, D_p = 2.5–10 μm) accumulate within the lower respiratory tract (Chen et al., 2015a). Fine particles ($PM_{2.5}$, D_p = 0.1–2.5 μm) can penetrate the alveoli of the lungs in the lower respiratory tract (Weber et al., 2014) and ultrafine particles ($PM_{0.1}$, D_p < 0.1 μm) are small enough to enter the blood stream and adversely affect the circulatory systems (Popek et al., 2015; Weber et al., 2014). Prolonged exposure to PM has resulted in close to a million premature deaths annually and is ranked thirteenth among the most common causes of mortality worldwide (WHO, 2002).

Urban greening has been increasingly explored as an effective way to remediate the negative impact of PM pollution in cities (Chen et al., 2016; Mo et al., 2015; Selmi et al., 2016). Among the common air pollutants, airborne PM has shown a higher potential to be reduced by plants (Yli-Pelkonen et al., 2017). Plants can remove PM from the air via dry deposition, a process that involves gravitational sedimentation, particle impaction, interception, as well as Brownian diffusion (Slinn, 1982). For a given area, plants are more effective at

Abbreviations: D_p, aerodynamic diameter; PM, particulate matter; $PM_{2.5}$, particulate matter of aerodynamic diameter 0.2–2.5 μm; PM_{10}, particulate matter of aerodynamic diameter 2.5–10 μm; PM_{100}, particulate matter of aerodynamic diameter 10–100 μm; ALA, average leaf area; SLA, specific leaf area

* Corresponding author.
Email addresses: chiamzy@u.nus.edu (Z.Y. Chiam); xp.song@u.nus.edu (X.P. Song); hughtan@nus.edu.sg (H.T.W. Tan)

https://doi.org/10.1016/j.scitotenv.2019.06.263
0048-9697/ © 2019.

UNCORRECTED PROOF

capturing PM from surrounding air than concrete surfaces (Dzierżanowski et al., 2011; Tallis et al., 2011). Multiple vegetation characteristics influence the overall effectiveness of plants and determine whether PM is retained on plant surfaces, accumulated within plant wax, re-suspended to the atmosphere or washed off by rain (Przybysz et al., 2014).

Numerous field studies have assessed the overall efficiency of PM capture across different plant species (Mo et al., 2015; Mori et al., 2015; Sæbø et al., 2012), and identified leaf traits that are associated with PM deposition (Grote et al., 2016; Liu et al., 2012; Weerakkody et al., 2018). On plant leaves, PM can be deposited on leaf surfaces (surface PM), or become embedded within the wax of leaves over time (in-wax PM) (Dzierżanowski et al., 2011). The amount of surface PM is influenced by various morphological and anatomical leaf traits (Liu et al., 2012; Räsänen et al., 2013). Generally, traits such as leaf hairiness, average leaf area and specific leaf area have shown to be strongly correlated with surface PM deposition (Prusty et al., 2005; Sæbø et al., 2012; Song et al., 2015). Over time, PM can also accumulate on leaves in the form of in-wax PM, affected by factors such as the thickness of the waxy layer and the amount of PM exposure over time (Dzierżanowski et al., 2011).

Despite the abundance of studies that examine PM deposition on plant leaves, variations in species selection, PM source, local environment conditions and spatiotemporal scope limit the comparability between results (Cai et al., 2017). Compared to assessments primarily based on species, those based on quantified plant traits can help to build generalizable conclusions applicable across broader scales and contexts. Several controlled experiments have explored the associations between PM deposition and leaf traits such as hairiness and specific leaf area (Sæbø et al., 2012; Weerakkody et al., 2018). However, important traits such as hairiness are often classified as ordinal or binary variables, and the extent to which each trait contributes to deposition remains unclear.

To fill these knowledge gaps, our study aims to investigate the effect of leaf traits on the dry deposition of PM on leaf surfaces within experimentally-controlled conditions. We adapted standardized protocols to quantify several key traits—hairiness, average leaf area (ALA) and specific leaf area (SLA)—and selected plants species that spanned a wide range of values for each trait. Most studies target plants species in temperate regions that may not necessarily grow outside of their local context (Nowak et al., 2014; Song et al., 2015; Yang et al., 2004). Thus, we focus on 20 tropical species commonly grown in Asia, where people tend to spend more time outdoors (Wong et al., 2008). Specifically, this study seeks to address the following questions:

1. What is the deposition rate on the leaves of each plant species, across varying PM size fractions ($PM_{2.5}$, PM_{10})?
2. How does hairiness on the upper (adaxial) and lower (abaxial) leaf surfaces contribute to overall deposition, across varying PM size fractions?
3. How do leaf hairiness, average leaf area and specific leaf area affect the deposition rate, across varying PM size fractions?

2. Methods

2.1. Plant material

Situated near the equator, Singapore is a city-state that experiences a year-round tropical climate. The city experiences air pollution that periodically exceeds the World Health Organization guidelines for PM_{10} and $PM_{2.5}$ (Singapore Department of Statistics, 2017; WHO, 2017; Yuen et al., 2012) and occasionally faces elevated levels of air pollution caused by unregulated burning of peat land in neighboring regions (Betha et al., 2014; Siao et al., 2007). Plant materials were collected from the National University of Singapore Kent Ridge Campus (1° 17′ N, 103° 46′ E) and Central Catchment Nature Reserve (1° 24′ N, 103° 47′ E). Twenty plant species across 16 families were selected to provide a wide variation in the values for three leaf traits: leaf hairiness, ALA and SLA (Table 1). For each species, three mature plants were sampled (60 individuals in total). From each individual, 400–500 cm^2 of mature leaves were collected for the wind-tunnel experiment. To preserve moisture and integrity, leaves were sealed in resealable plastic bags upon collection and immediately stored in the refrigerator upon returning to the laboratory. All leaves were processed in the laboratory within 24 h of collection.

2.2. Wind tunnel design

The experimental study was conducted by supplying a continuous stream of PM to a straight-duct wind tunnel from a smoke-emission chamber (Fig. 1). Incense sticks (Lemon and Pomelo Incense Sticks, JinWangLai; widely used in Singapore) were utilized as the source of

Table 1
Plant species and their mean leaf trait values in this study. Nomenclature follows that of Chong et al. (2009).

Species	Family	SLA ($cm^2 g^{-1}$)	Top H' (mm^2)	Btm H' (mm^2)	ALA (cm^2)
Adinandra dumosa Jack	Pentaphylacaceae	119.37	0	0	46.15
Bauhinia semibifida Roxb.	Fabaceae	202.58	2693	1794	42.80
Bridelia stipularis (L.) Blume	Phyllanthaceae	192.07	280	992	47.03
Bridelia tomentosa Blume	Phyllanthaceae	191.78	101	737	29.22
Callicarpa longifolia Lam.	Lamiaceae	330.19	652	409	72.22
Cheilocostus speciosus (J. Koenig) C.D. Specht	Costaceae	274.71	152	2288	48.65
Clidemia hirta (L.) D. Don	Melastomataceae	190.14	52	65	41.72
Ficus aurata (Miq.) Miq.	Moraceae	172.31	327	1580	65.96
Gardenia carinata Wall. ex Roxb.	Rubiaceae	144.63	189	380	74.03
Gironniera nervosa Planch.	Cannabaceae	179.28	109	401	64.81
Gnetum gnemon L.	Gnetaceae	177.02	0	0	42.10
Grewia laevigata Vahl	Malvaceae	212.81	98	164	38.35
Melastoma malabathricum L.	Melastomataceae	211.34	103	285	26.39
Muntingia calabura L.	Muntingiaceae	221.34	1282	2900	33.74
Piper aduncum L.	Piperaceae	324.51	190	1164	78.15
Spathodea campanulata P.Beauv	Bignoniaceae	203.50	80	134	65.11
Strobilanthes crispa (L.) Blume	Acanthaceae	166.84	253	72	74.67
Syzygium myrtifolium Walp.	Myrtaceae	128.40	0	0	12.01
Timonius wallichianus (Korth.) Valeton	Rubiaceae	278.78	809	5839	71.23
Vitex pinnata L.	Lamiaceae	183.09	188	243	48.38

UNCORRECTED PROOF

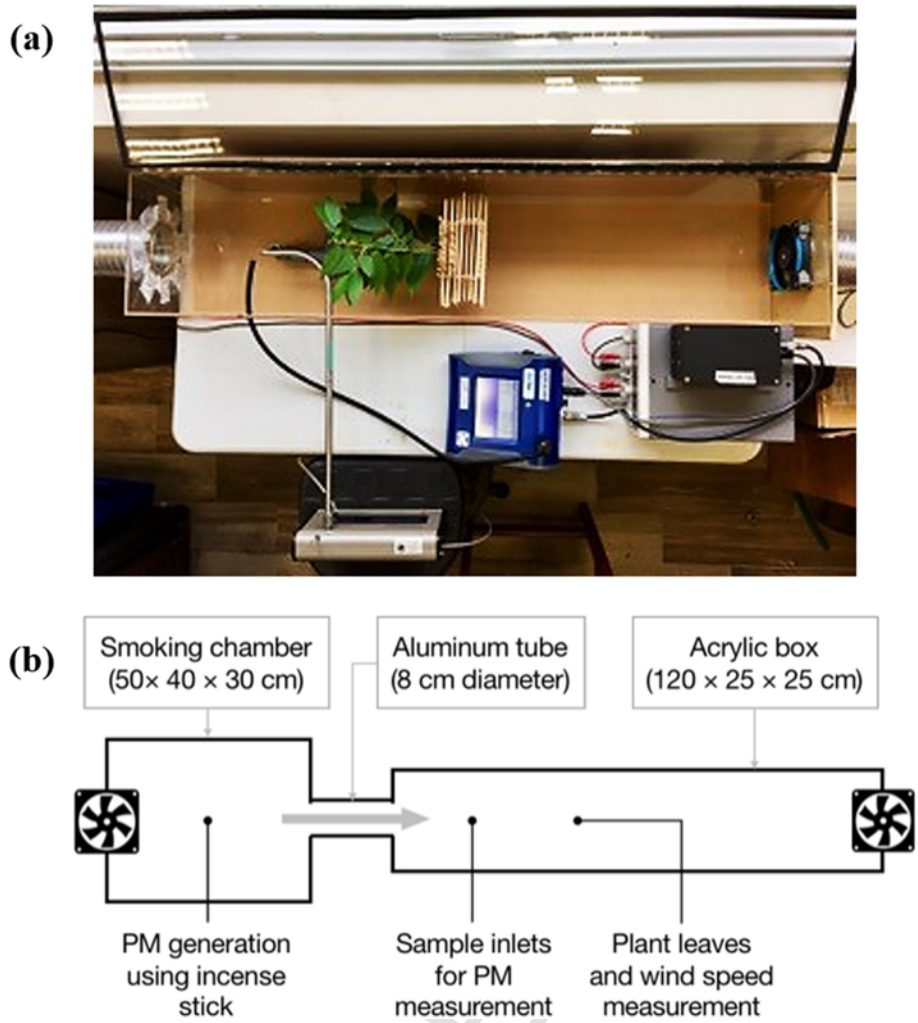

Fig. 1. (a) Top view and (b) schematic diagram of the custom-made, air-flow chambers. The grey arrow indicates the direction of the air flow, from the smoke-emission chamber (not shown in top-view diagram) to the wind tunnel. PM was generated using a single incense stick. PM count and mass concentrations were measured using a Mini-Laser Aerosol Spectrometer. Branches were positioned planar to the air flow and wind speed measurements were taken right before contact with the leaves.

PM in the smoke chamber. Incense burning is an important tradition in many Asian nations and emissions from incense closely resemble those from burning wood, since the raw materials of incense sticks are mainly wood powder and bamboo (Chang et al., 2007; Yang et al., 2007). Incense emissions include particles that are both soluble and insoluble in water; the soluble and insoluble fractions are brand-specific as they are influenced by components such as colour additives and fragrance oils (Li and Hopke, 1993). The use of incense is favourable as particles produced show low hydroscopic growth (Vu et al., 2017), and studies have found that particle diameters were relatively consistent under varying levels of humidity (Chang et al., 2007). While the PM generated may not be similar in the chemical content or health risks associated with other pollutant sources, incense generally produces a wide variation in particle sizes (ranging from 0.1 μm to 10 μm) (Lin et al., 2015; Lung and Hu, 2003), which is comparable to traffic-derived PM (Pant and Harrison, 2013).

Airflow was generated by two high performance axial fans (Air Series SP120 High Performance, Corsair) that were attached to both ends of the setup (Fig. 3). Wind speeds were regulated at 2.2 m s^{-1} using an adjustable power source (72-8690A, Tenma, UK) and monitored with a multi-purpose anemometer (45,170, Extech, US) to produce the required air flow through the tunnel. A turbulent air flow (Reynolds number > 4000) was used to mimic the wind conditions of the natural environment as in other studies (Lin and Khlystov, 2011; Räsänen et al., 2012) and the pressure drop throughout the whole setup was negligible. Air temperature and relative humidity were also measured within the wind tunnel, while the PM count and mass concentration were measured with a Mini-Laser Aerosol Spectrometer 11-R fitted with an isokinetic duct probe 1152 (GRIMM, Ainring, Germany). Readings were taken at 6-s intervals. The concentrations of $PM_{2.5}$ and PM_{10} in the wind tunnel ranged from 67.00 ± 1.00 μm m^{-3} (mean ± S.E) and 11.90 ± 1.65 μm m^{-3} respectively. These were higher than ambient concentrations in Singapore, which averaged 17 μm m^{-3}/year for $PM_{2.5}$ and 27 μm m^{-3}/year for PM_{10} from 2011 to 2017 (Singapore Department of Statistics, 2018). This was done to simulate conditions where PM removal from a more polluted atmosphere becomes increasingly important, and to ensure sufficient deposition of PM during the experiment to be above the detection threshold of our microbalance. PM_{100} was excluded from the analysis, owing to its relatively lower health risk and lower concentrations produced during the experiment (see Appendix A for more information).

2.3. Measuring dry deposition

This study assessed short-term surface deposition on plant leaves; the long-term accumulation of PM in the waxy layer of leaves was not within the scope of our analyses, given the short duration of the wind tunnel simulations. While other observational studies measure the deposition of both types of PM in situ (Dzierżanowski et al., 2011; Sæbø et al., 2012), extracting in-wax PM from our leaf samples would introduce particles that were deposited under different environmental conditions, outside the time period of our experiment. Thus, even though waxiness is an important trait for long-term PM accumulation, we focus on leaf traits that affect short-term surface deposition. A focus on surface deposition also better represents the flux of PM resulting from short-term deposition and removal via natural processes (i.e. washing off by frequent rain in the tropics). The following protocol to extract surface PM was based on the combined methods of two studies (Dzierżanowski et al., 2011; Sæbø et al., 2012).

Prior to PM exposure, leaves of each batch (400–500 cm^2) were washed thoroughly with deionised water. A non-hair loss paint brush with bristles made of 100% polyester filament (Stanley, UK) was used to aid the removal of PM adhering to leaf surfaces; this method of washing the leaves is known to be effective in removing almost all surface PM (Song et al., 2015). Leaves were left to dry in a laminar flow cabinet (Esco Micro, Singapore) before they were placed in the wind tunnel. The use of the laminar flow cabinet allowed the leaves to dry quickly within an uncontaminated environment. Prior to PM exposure, filter papers were also placed in a drying oven (MB6, WTC Binder, Germany) for 48 h at 60 °C. They were subsequently taken out and left at a relative humidity of 65% for 24 h, before weighing them on a high precision electronic balance (ABT 100-5 M, KERN, Germany). A pilot study found that re-humidification for at least 13 h was required for the weight of filter papers to stabilize (Fig. B.1, Appendix B).

Each batch of leaves was exposed to PM generated by a stick of incense for a duration of 15 min at a wind speed of 2.2 $m\,s^{-1}$. After exposure to PM, leaf petioles were removed. Leaves were then washed with 300 ml of deionised water for three minutes to remove all surface PM, with the aid of the non-hair loss brush. The water was then subjected to multiple levels of filtration within the laminar flow cabinet. First, the water was filtered with a 100 μm metal sieve to remove particles larger than 100 μm. Using a 47 mm plastic filter funnel with stopper support assembly attached to a vacuum pump (R-300, Boeco, Germany), the filtrate was filtered consecutively using Type 91, Type 42 and PTFE membranes (all Whatman, UK). This corresponded to the removal of PM_{100}, PM_{10} and $PM_{2.5}$ from the filtrate, respectively. The dry weight of each filter paper was measured after drying and humidification. The weight of PM in each fraction was obtained as the difference between the pre- and post-experiment weights of the filter papers. It should be noted that this method only accounts for insoluble PM deposited onto leaf surfaces. Soluble PM is rarely assessed in recent studies (Chen et al., 2017, 2016), though it may be considered in future works by performing ion chromatography or spectrometry on the filtrate (Beckett et al., 1995). We therefore interpret our results as the amount of physical, insoluble PM deposited on leaf surfaces.

2.4. Leaf traits

Owing to the unequal distribution of hairs on leaves, leaf surfaces were classified into three parts: (1) primary veins, (2) secondary veins and (3) the "lamina" which included very fine, higher order veins (Fig. 2). For each part, three photographs of a small area (~1 mm^2) were taken under a compound microscope (M205C, Leica Camera, Wetzlar, Germany) and the number of hairs per unit area (mm^{-2}) were counted. The total leaf hairiness (H; units in mm^{-2}) of each leaf was then estimated as the sum of primary vein hairiness (H_P), secondary vein hairiness (H_S) and lamina hairiness (H_L), each weighted by the percentage area of the respective anatomical part:

$$H = \frac{A_P}{A_T}H_P + \frac{A_S}{A_T}H_S + \frac{A_L}{A_T}H_L$$

where A_P, A_S, A_L and A_T denote the area of primary vein, secondary vein, lamina and leaf total, respectively. The total leaf hairiness H was estimated for both the upper (H_T) and lower (H_B) surface. For each species, the entire procedure was repeated on three mature leaves. (Table 2).

To determine the area (A) of each anatomical part, five leaves for each species were bleached and stained using safranin O solution following well-established protocols from both Pérez-Harguindeguy et al. (2013) and Blonder and Enquist (2014). Stained leaves were later scanned and analysed using the software ImageJ (Collins, 2007) to obtain average area of A_P, A_S, A_L and A_T.

Both the ALA and SLA were calculated for every batch of leaves used in the wind-tunnel experiment. After the final wash, the leaves

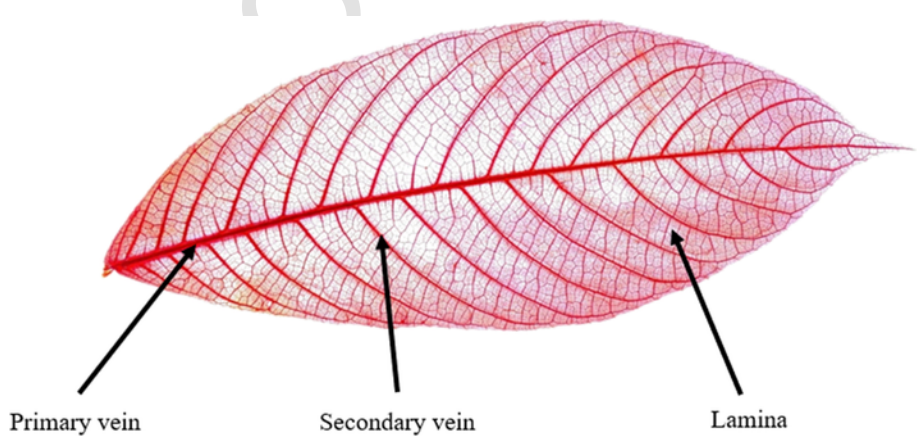

Fig. 2. An example showing a leaf of *Gironniera nervosa* after the bleaching and staining procedure.

UNCORRECTED PROOF

Table 2
Leaf traits measured in this study. (Hair = leaf hairiness; ALA = average leaf area; SLA = specific leaf area).

S/No.	Leaf trait (unit)	Definition	Formula
1	Hair (no. mm^{-2})	Number of hairs per unit area (for both upper and lower leaf surfaces)	$\left(\frac{\text{Number of hairs}}{\text{Measured area}}\right)$
2	ALA (cm^2)	Average leaf area per batch	$\left(\frac{\text{Total batch leaf area}}{\text{Number of leaves}}\right)$
3	SLA ($cm^2\,g^{-1}$)	Ratio of leaf area to dry weight per batch	$\left(\frac{\text{One sided leaf area}}{\text{Dry weight}}\right)$

were scanned (Scanjet G4050, HP) to obtain the total leaf area per batch, before drying in an oven for one week at 60 °C. Dry leaves of each batch were subsequently weighed using an electronic balance (MS304TS, Mettler Toledo, Switzerland). ALA was determined by dividing the total leaf area by the number of leaves per batch, while SLA was calculated by dividing the total leaf area of each batch by their dry weight (Table 2).

2.5. Statistical analyses

The relative deposition velocity (Dv_{rel}, $m\,s^{-1}$) of each plant species was derived in order to account for the cumulative fluxes of PM input on the leaves, and thus the slight variations in PM exposure between replicates (Lin et al., 1994). This was calculated as:

$$\mathrm{Dv_{rel}} = \mathrm{M/C}$$

where M ($\mu g\,m^{-2}\,s^{-1}$) is the weight of PM deposited on the leaves per unit area per time and C ($\mu g\,m^{-3}$) refers to the pollutant concentration measured within the wind tunnel (see Litschike and Kuttler (2008) and Nowak et al. (2014) for a more detailed explanation).

A study conducted by Hwang et al. (2011) reported that the upper surface of leaves had a disproportionately higher contribution to the capture of PM particles. For each PM fraction, the ratio of effective leaf hairiness, H', was thus estimated for the upper and lower surfaces, each weighted by a relative contribution factor p that ranges from 0 to 1:

$$H' = p_H H_U + \left(1 - p_H\right) H_L$$

where:

H' = effective leaf hairiness

H_U = total leaf hairiness on the upper surface

H_L = total leaf hairiness on the lower surface

p_H = relative contribution factor of leaf hairiness on the upper surface from 0, 0.01, …, 0.99, 1.0

For each PM faction, the deposition velocity (Dv) was modelled as a function of the three leaf traits (Table 2) in a linear regression:

$$Dv \sim H' + ALA + SLA.$$

Multiple models with all possible combinations of p_H for the H' variable were fitted and ranked according to Aikake's Information Criteria corrected for small sample size (AICc), in order to estimate the relative contribution of hairiness on the upper surface of leaves (p_H). These values were used for subsequent analysis.

Collinearity between variables was tested using the 'vif' function in the R package 'car' and no strong multicollinearity was found (Fox and Weisberg, 2011). The response variable (Dv) was log-transformed and leaf hairiness was square root-transformed to meet the linear model's assumption of normality. All explanatory variables were scaled to zero mean and unit standard deviation prior to analysis. One of the three individuals for the plant species *Vitex pinnata* was removed ($PM_{2.5}$), owing to its high leverage based on Cook's distance. Model selection was performed using the function 'model.sel' from the package 'MuMIn'. Models within 2 units of AICc from the top-ranked model were shortlisted (Burnham and Anderson, 2002). All analyses were conducted in R 3.4.4 (R Core Team, 2017).

3. Results

3.1. Species differences in dry deposition

For each PM fraction, there was a wide variation in their deposition velocities across the 20 plant species (Fig. 3). On average, the leaves of *Grewia laevigata* (mean ± S.E = 2.75 ± 0.17 $m\,s^{-1}$), *Bauhinia semibifida* (2.34 ± 0.19 $m\,s^{-1}$) and *Ficus aurata* (2.25 ± 0.07 $m\,s^{-1}$) had high rates of PM_{10} deposition. *Muntingia calabura* (0.26 ± 0.01 $m\,s^{-1}$) which had the highest rate of $PM_{2.5}$, also had similarly high deposition velocities for PM_{10}. Species such as *Syzygium myritifolium* and *Adinandra dumosa* with glabrous (smooth and hairless) leaf surfaces had low deposition velocities across both PM fractions.

3.2. Interactions between hairiness and different leaf surfaces

Hairiness on the upper and lower surfaces of leaves did not contribute equally to PM deposition and their relative contributions varied across the different fractions of PM (Table C.1, Appendix C). Generally, the upper surface played a more important role in the deposition of PM—it accounted for 78% and 64% contribution to the effective hairiness that influenced the deposition of $PM_{2.5}$ and PM_{10}, respectively.

3.3. Leaf traits influencing dry deposition

In general, SLA showed a negative relationship with the deposition velocity while hairiness had a positive relationship (Fig. 4) and the trend was consistent throughout both $PM_{2.5}$ and PM_{10} (Fig. 4). On the other hand, ALA showed no observable relationship with both $PM_{2.5}$ and PM_{10}. Although SLA and leaf hairiness were important predictors for both $PM_{2.5}$ and PM_{10}, only leaf hairiness showed strong positive effects for both. SLA showed strong effects for PM_{10} but for $PM_{2.5}$ the effect was less obvious (as indicated by their overlap with zero, Fig. 4). Across the other candidate models within 2 ΔAICc (Table D.1, Appendix D), SLA and ALA generally had a weak or no effect on deposition velocity of $PM_{2.5}$ and PM_{10}. Lastly, the top ranking models explained ~16% and ~21% of the variation observed in the deposition velocity of $PM_{2.5}$ and PM_{10}, respectively.

4. Discussion

4.1. Effectiveness of different tropical plant species

While plants have the potential to reduce levels of atmospheric PM, one key consideration is to understand their effect on different PM size fractions. The comparison between plant species revealed two distinctive trends. Species with a greater PM deposition tend to have a higher hair density, including *Ficus aurata*, *Muntingia calabura*, *Grewia laevigata* and *Bauhinia semibifida*. On the other end of the spectrum, less effective species such as *Syzygium myrtifolium*

UNCORRECTED PROOF

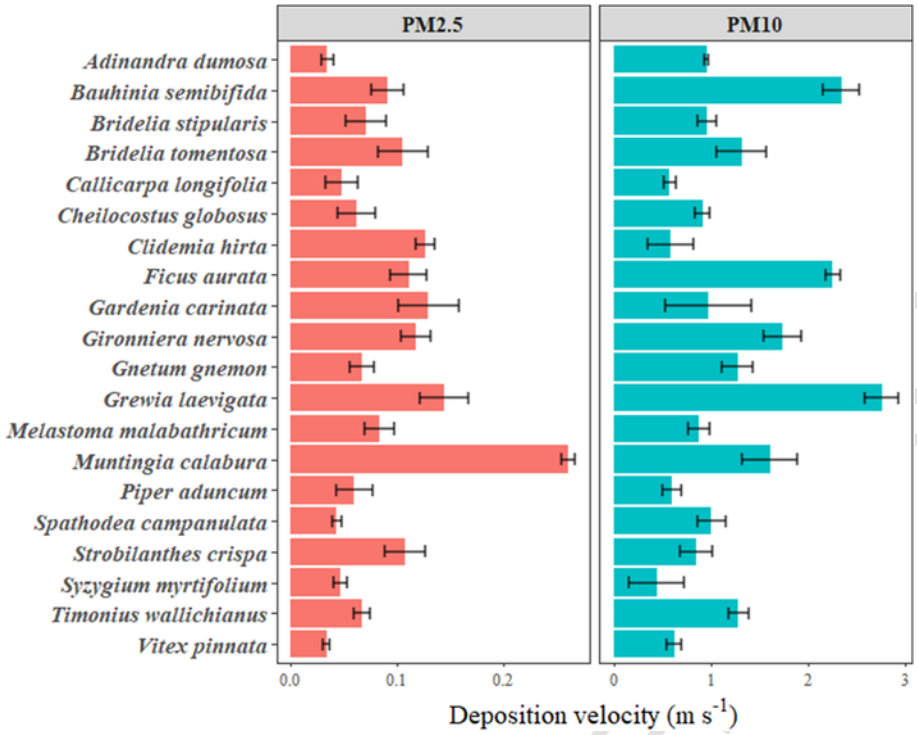

Fig. 3. The average deposition velocities of $PM_{2.5}$ and PM_{10} on leaf surfaces of the 20 tropical plant species in this study. Error bars denote 1 standard error from the mean.

and *Adinandra dumosa* tend to have few or no hairs on their leaves. Interestingly, *Gnetum gnemon* with glabrous surfaces had relatively high deposition velocities for PM_{10} as compared to other hairy species. This unexpected result could have been due to the cup-shaped structure or grooves on the leaf surfaces of this species, which were observable under the microscope. Using synthetic leaves, Weerakkody et al. (2018) have shown that the physical structure of whole leaves also plays a critical role in the PM deposition, owing to its influence on both PM retention and resuspension back into the atmosphere.

Overall, *Muntingia calabura* was able to accumulate PM of both size fractions effectively. Apart from its hairy leaves, the sticky texture of its leaf surfaces may contribute to PM deposition and retention. It is a fast-growing tree that has high tolerance for poor soil conditions, which has enabled it to thrive within the urban landscape. With little to no information documented about plants suitable for pollution mitigation in the tropics, this species could be a potential candidate. On the other hand, *Syzygium myrtifolium*—a popular hedge plant planted in the tropics—was less effective in PM accumulation on leaf surfaces. Its smooth and glabrous surfaces showed low deposition velocities across all PM size fractions, which suggests the importance of micromorphological structures for PM deposition. In order to combat elevated levels of air pollution, similar assessments of other commonly planted species within the tropics can help establish the overall effectiveness of species currently used in urban landscaping.

4.2. Contribution of different leaf surfaces toward dry deposition

Leaf hairs on upper and lower surfaces of leaves may contribute differently toward surface deposition. By quantitatively examining the interactions between hairiness and different leaf surfaces, we can understand the extent to which hairiness on each surface contributes toward PM deposition. Generally, results show higher contribution of the upper surface toward deposition of all PM fractions, which is consistent with results of the study by Wang et al. (2006). This is not surprising, owing to the effect of gravity on particles. Since the PM size fractions tested were larger than 0.1 μm, the effect of Brownian motion—the random movement of particles—on PM deposition is less important compared to the effect of gravitational sedimentation (Chen et al., 2015; Cai et al., 2017). It is likely that particles adhering to the lower surface could also re-suspend owing to gravitational forces and air turbulence. Thus, species that are hairy primarily on lower leaf surfaces, such as *Timonius wallichianus*, may be less effective in PM deposition for these larger size fractions.

4.3. Analysis by leaf traits for phytoremediation studies

The screening of new plant species for phytoremediation is an important part of research to ameliorate the effects of air pollution. On the other hand, analyses that quantify the contributing effects of leaf traits can help us understand underlying mechanisms of PM deposition, and estimate spatially-averaged deposition velocities across areas containing different plant species. This can improve the efficiency of decision making especially in the tropics, where there is a high diversity of plant species to choose from. The importance of two leaf traits—leaf hairiness, SLA—were reflected by their effects on the deposition velocity of $PM_{2.5}$ and PM_{10} (Fig. 4). Both traits were consistent in their prediction of the effect direction across the PM size fractions (Fig. 4), which make them suitable to be used for rapid assessments of species suitability for PM deposition.

The positive association between leaf hairiness and deposition velocity was similar to results obtained from previous studies (Cai et al., 2017; Dzierzanowski and Gawronski, 2011; Sæbø et al., 2012). However, a minimum hair density may have to be reached before it significantly contributes toward increased PM deposition (Chen et al., 2017). This could be a possible reason why species with mildly hairy leaves such as *Vitex pinnata* and *Spathodea campanulata* had

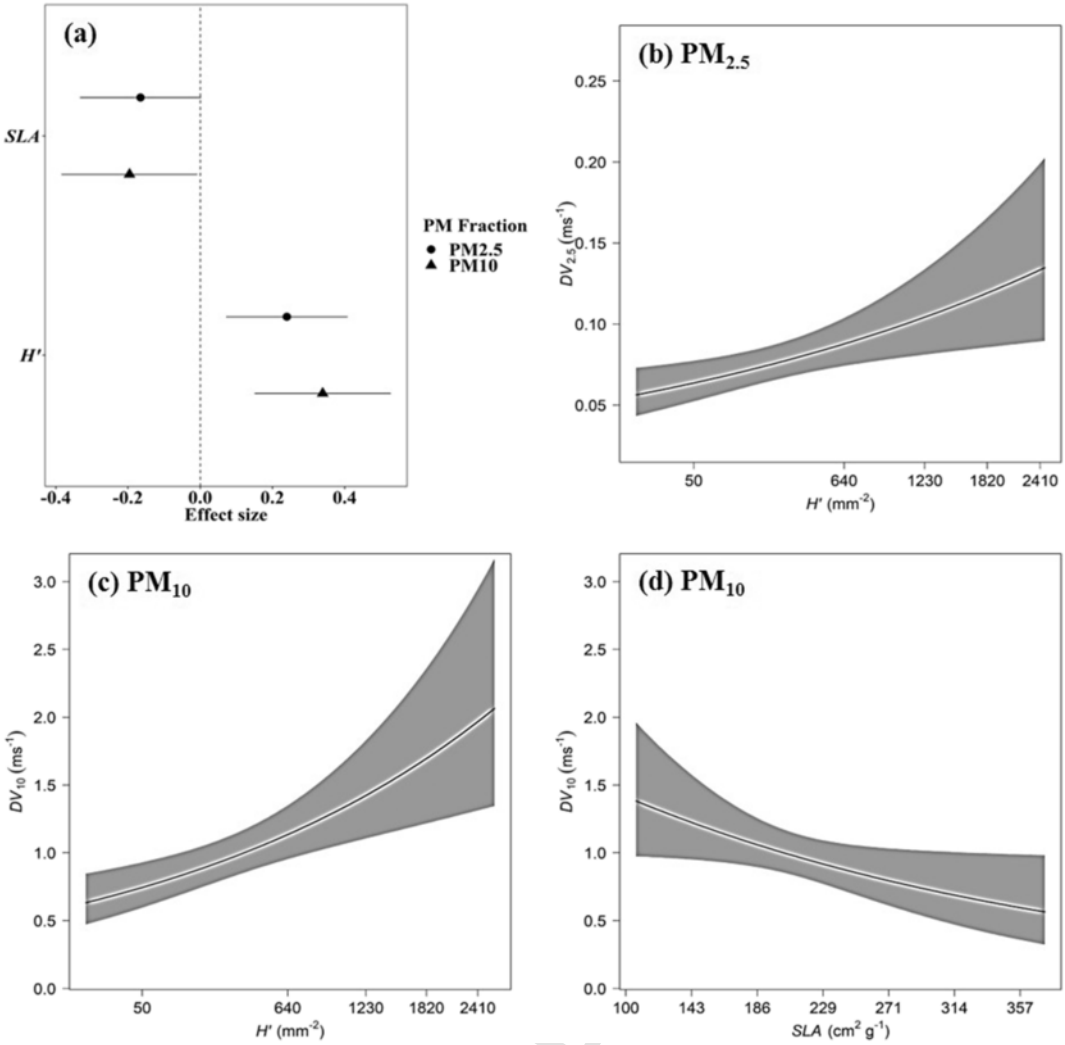

Fig. 4. Final model results presented as (a) a coefficient plot of $PM_{2.5}$ and PM_{10} deposition velocities, and as model predictions for deposition velocities of: (b) $PM_{2.5}$ and (c) PM_{10} with increasing leaf hairiness; and (d) PM_{10} with varying specific leaf area. Symbols in (a) denote standardized effect sizes and the bars represent the 95% confidence intervals; confidence intervals that do not overlap with the zero-value (vertical dashed line) indicate a strong effect on the response. In (b) to (d), black lines denote fitted regression lines with shaded regions representing the confidence intervals (95%). Abbreviations: *SLA* = specific leaf area; *H′* = leaf hairiness.

deposition velocities that were similar to the smooth glabrous leaves of *Syzygium myrtifolium*. From this study, it is also worth noting that *Cheilocostus speciosus* had a very high hair density (~152 hairs mm^{-2}) on the upper surfaces of leaves but showed a relatively low deposition rate. This may be attributed to leaf hairs being long, straight and aligned, resulting in a smooth and silky surface texture. Another interesting observation involved the high deposition velocity of *Clidemia hirta* for $PM_{2.5}$. Although *Clidemia hirta* had one of the lowest hair densities, its unusually long hairs may have increased the effective area of contact with the circulating PM in the wind tunnel. Hence, apart from hair density, the length, type and direction of hairs may be possible confounding factors that influence PM deposition. Another influencing factor could have been the water repellence of leaves that had high hair densities, thus preventing the removal of particles during the washing procedure. Although caution was exercised to ensure that the washing of the leaves was thorough, perhaps a more rigorous method of washing can be used. Future studies may consider using ultrasonic waves to better remove the PM on leaf surfaces (Liu et al., 2012).

A larger ALA may contribute to a larger boundary layer around leaf surfaces, that may result in its negative correlation to PM deposition (Chen et al., 2017; Song et al., 2015; Weerakkody et al., 2018). This boundary layer refers to the still air surrounding the leaf (Burkhardt, 2010) which increases its resistance to the deposition of particles (Weerakkody et al., 2018). However, our results showed no observable relationship between ALA and PM deposition.

SLA is an important trait to consider at both the scale of individual plants (Chen et al., 2015a), as well as individual leaves as shown in this study. At the scale of individual plants, a high SLA is often associated with larger and thinner leaves (Reich et al., 1991), resulting in a higher degree of overlap (Chen et al., 2015a) that reduces the effective area for deposition. At the scale of individual leaves, the effects of SLA on PM deposition still lack a mechanistic explanation. It is possible that leaves with a lower SLA invest more into structural stability (Burgess and Dawson, 2007). These stiffer leaves might have swayed less under the wind in the experimental chamber, thus reducing the resuspension of PM back into the air. However, as the actual movement of leaves was not quantified during the experiment,

future studies are needed to validate the possibility of such an effect on deposition.

4.4. Scaling up to plant-level PM accumulation

Studies that examine leaf-level PM deposition can help us build a mechanistic understanding of PM accumulation in plants. However, there is also a need to scale up analyses to the level of entire plants, since each species has different plant-level traits that may influence the rate of PM deposition. Varying crown morphologies, growth architecture, as well as the density and spatial distribution of foliage can each exert a positive or negative impact on overall PM deposition (Przybysz et al., 2014). For instance, depending on the relative influence of different plant traits, a hairy-leafed species with a sparse crown may not be as effective as a species with non-hairy leaves but a densely-packed crown.

Studies of plant-level PM accumulation often scale up leaf-level measurements across the total leaf area, which assumes similar deposition rates across all parts of the tree. For example, the leaf area index (LAI) of the tree canopy has been used to estimate the total foliage area, which was then multiplied by the masses of different PM fractions measured per specific area (Song et al., 2015). Software such as i-Tree Eco (n.d.) have also been developed to estimate the deposition of PM on trees across large spatial scales, based on the canopy projection area (Hirabayashi et al., 2015; Song et al., 2018). However, to date, i-Tree has only been adapted for the countries outside the Asian region (USDA, n.d.). More work is needed to gather knowledge on the vast diversity of plant species in the tropics—such locations differ from temperate regions in their climate, ecology, and human factors that influence people's exposure to air pollution (Song et al., 2017). Furthermore, there is a need to consider that some assumptions may not hold true when estimating plant-level PM deposition. The use of LAI to scale leaf-level deposition to entire trees may only be applicable for trees that have sparse crowns. In a study conducted by Hofman et al. (2014), authors revealed that as leaf density increases, the amount of dust per unit area deposited on leaves decreases. This negative relationship between deposition and leaf density may be due to higher deposition along the edges of the tree crown, as well as a closer proximity to the PM source. For such a scenario, it might be better to make use of ground-based LiDAR as per Hofman et al. (2014) which can provide a detailed representation of a tree's three-dimensional structure. Such an approach may be able to account for the reduced deposition of PM with increasing leaf density and provide a more accurate evaluation of plant suitability for pollutant mitigation.

5. Conclusions

This study examined three key leaf traits and their effect on surface deposition of $PM_{2.5}$ and PM_{10}, under experimentally-simulated conditions. At the scale of individual leaves, the species *Grewia laevigata*, *Ficus aurata* and *Muntingia calabura* were effective in capturing most PM size fractions. Analysis by leaf traits showed that SLA and leaf hairiness were negatively and positively correlated, respectively, to deposition for both PM size fractions. Our findings show that both the physical structure of leaves and their interactions with different leaf surfaces can influence PM deposition. Species suitability for pollution mitigation should thus consider these interactions and their contribution to PM accumulation, across multiple spatial scales.

CRediT authorship contribution statement

Zhong Yu Chiam: Conceptualization, Methodology, Investigation, Formal analysis. **Xiao Ping Song:** Conceptualization, Methodology, Resources, Funding acquisition, Writing - review & editing. **Hao Ran Lai:** Conceptualization, Methodology, Formal analysis, Writing - review & editing. **Hugh Tiang Wah Tan:** Supervision, Project administration, Funding acquisition, Writing - review & editing.

Declaration of Competing Interest

The authors declare that there is no conflict of interest.

Acknowledgements

The authors would like to thank the Office of Facilities Management of the National University of Singapore and the National Parks Board (Permit number NP/RP16-012-1) for permission to collect plant material at field sites. We are also grateful to Michiel van Breugel and David Cheong Kok Wai for the use of equipment. This research did not receive any specific grant from funding agencies in the public, commercial or not-for-profit sectors.

Appendix A. Concentrations of particulate matter produced by incense

Incense sticks (Lemon and Pomelo Incense Sticks, JinWangLai; widely used in Singapore) were used to generate PM for the wind tunnel experiment. For each replicate, one stick of incense was burned for 15 min and the concentration of PM within each size fraction was measured using a Mini-Laser Aerosol Spectrometer 11-R fitted with an isokinetic duct probe 1152 (GRIMM, Ainring, Germany). Based on the results of 60 replicates, the incense used produced higher amounts of $PM_{2.5}$ and PM_{10}. While PM_{100} was also produced, the low concentrations would have required more replicates to establish significant differences and had the potential to compromise the accuracy of the results. Hence, the ability of plants to filter PM_{100} was excluded from our study.

UNCORRECTED PROOF

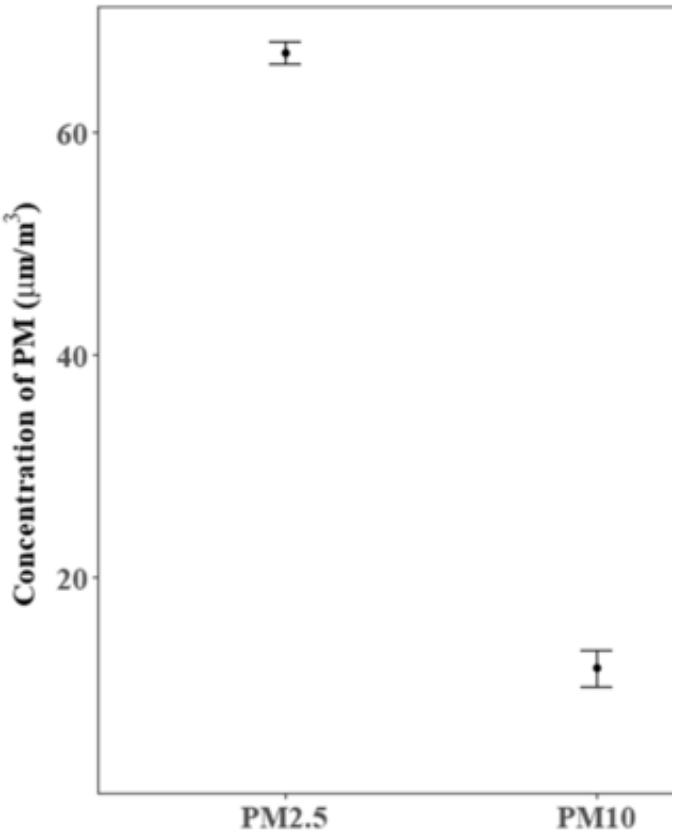

Fig. A.1. Average concentration of $PM_{2.5}$, PM_{10} and PM_{100} produced by a single incense stick. The filled circles represent the mean concentration of each PM fraction and error bars represent 1 standard error from the mean value.

Appendix B. Pilot study: duration for the re-humidification of filter papers

After removing the filter papers from the drying oven, the duration for their re-humidification is expected to be longer under humid tropical conditions. Based on pre-existing protocols, the duration for filter papers to stabilize and achieve a constant weight was insufficient. Results from our pilot study showed that weights continued to increase gradually. To resolve this problem, the three grades of filter papers (Type 91, Type 42 and PTFE membrane) were first dried for 48 h in a drying oven at 70 °C. They were subsequently taken out and left at 65% relative humidity for 24 h, before taking the first measurements on a high precision electronic balance. Deionised water was subsequently used to wet the filter papers, before placing them back in the drying oven for 48 h. Again, the filter papers were taken out of the oven and left to acclimatize for 4 h at ambient humidity levels. They were weighed every 30 min (from the fourth) till the eleventh hour. The last reading was performed after 24 h. Twenty-three measurements were made for each type of filter paper. The duration for the re-humidification of Type 91 filter paper (for PM_{100}) was included as it may potentially benefit future studies that consider this PM size fraction.

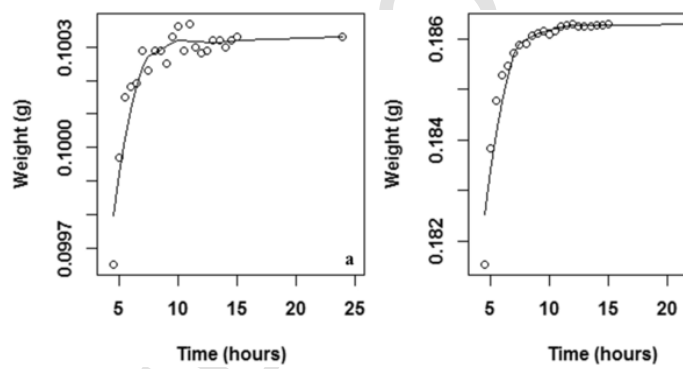

Fig. B.1. *Re*-humidification of the (a) PTFE membrane; (b) Type 42 filter paper (10 μm); and (c) Type 91 filter paper at 65% relative humidity.

Appendix C. Contribution of upper- and lower-surface hairiness of leaves

We modelled the deposition velocities of each PM fraction as a function of the three leaf traits (SLA, Hair and ALA). For each PM fraction, 100 models each with differing effective leaf hairiness—with the contribution factor ranging from 0 to 1—was estimated for the upper and lower surface. Based on Aikake's Information Criteria corrected for small sample sizes (AICc), the final models show that the upper surface had a higher contribution to PM deposition (Table C.1 and Fig. C.1).

Table C.1 The relative contribution of leaf hairiness for each leaf surface, on the deposition velocity of three PM size fractions. All species were included in the model.

Leaf surface	$PM_{2.5}$	PM_{10}
Adaxial (upper)	78%	64%
Abaxial (lower)	22%	36%

UNCORRECTED PROOF

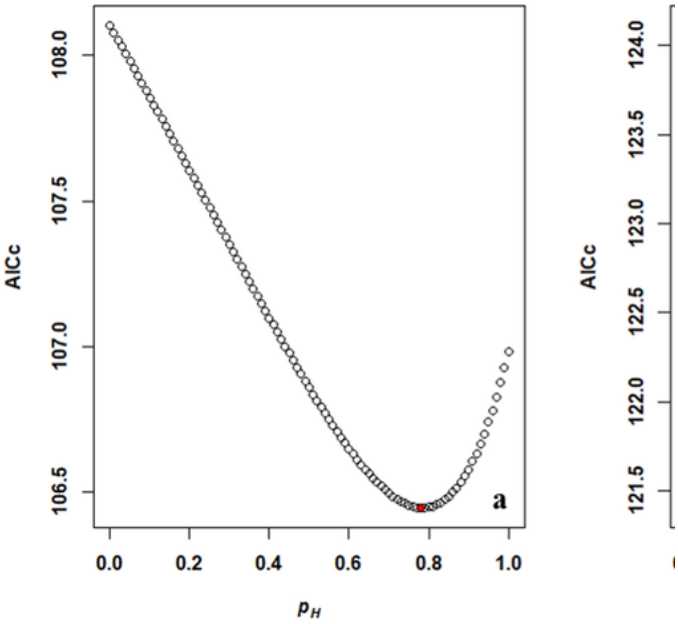

Fig. C.1. Changes in the Akaike Information Criteria corrected for small samples (AICc) of the candidate models for (a) $PM_{2.5}$; and (b) PM_{10} based on the relative contribution of hairiness on the upper surfaces of leaves (p_H). Open circles represent the different p_H values modelled. The filled circle that is red in colour represents the p_H that had the lowest AICc value (final model used for subsequent analysis).

Appendix D. Best models for each PM fraction

Each PM fraction was modelled against three traits: Specific leaf area, leaf hairiness and average leaf area. These models were ranked via Akaike Information Criteria corrected for small samples (AICc) and models within 2 ΔAICc from the top models were shortlisted.

Table D.1 List of models with ΔAICc < 2 for the three PM size fractions, ranked in order of Akaike weights. Slope estimate values in bold denote where the 95% CI of the averaged parameter estimates do not include zero. Abbreviations: $DV_PM_{2.5}$ = deposition velocity of $PM_{2.5}$; DV_PM_{10} = deposition velocity of PM_{10}; SLA = specific leaf area; H′ = effective leaf hairiness; ALA = average leaf area.

Model no.	Response variable	Explanatory variable	Slope estimates	SE	df	logLik	AIC
1	log($DV_PM_{2.5}$)	*SLA*	−0.165	0.084	4	−48.007	104.
		H′	**0.240**	0.084			
		H′	**0.161**	0.075	3	−49.983	106.
2		*SLA*	−0.147	0.087	5	−47.657	106.
	H′		**0.245**	0.087			
3		*ALA*	−0.063	0.078			
4	log(DV_PM_{10})	*SLA*	**−0.196**	0.094	4	−55.172	119.
		H′	**0.339**	0.094			

References

Beckett, P.K., Freer-smith, P., Taylor, G., 1995. Effective tree species for local air-quality management. J. Arboric. 26 (1).

Betha, R., Behera, S.N., Balasubramanian, R., 2014. 2013 Southeast Asian smoke haze: fractionation of particulate-bound elements and associated health risk. Environ. Sci. Technol. 48, 4327–4335.

Blonder, B., Enquist, B.J., 2014. Inferring climate from angiosperm leaf venation networks. New Phytol. 204, 116–126.

Burgess, S.S.O., Dawson, T.E., 2007. Predicting the limits to tree height using statistical regression of leaf traits. New Phytol. 174, 626–636.

Burkhardt, J., 2010. Hygroscopic particles on leaves: nutrients or desiccants?. Ecol. Monogr. 80, 369–399.

Burnham, K.P., Anderson, D.R., 2002. Model selection and multimodel inference: a practical information-theoretic approach (2nd ed). Ecol. Model.

Cai, M., Xin, Z., Yu, X., 2017. Spatio-temporal variations in PM leaf deposition: a meta-analysis *. Environ. Pollut. 231, 207–218.

Chang, Y.C., Lee, H.W., Tseng, H.H., 2007. The formation of incense smoke. J. Aerosol Sci. 38, 39–51.

Chen, J., Yu, X., Sun, F., Lun, X., Fu, Y., Jia, G., Zhang, Z., Liu, X., Mo, L., Bi, H., 2015. The concentrations and reduction of airborne particulate matter (PM10, PM2.5, PM1) at shelterbelt site in Beijing. Atmosphere 6 (5), 650–676.

Chen, X., Zhou, Z., Teng, M., Wang, P., Zhou, L., 2015. Accumulation of three different sizes of particulate matter on plant leaf surfaces: effect on leaf traits. Arch. Biol. Sci. 67, 1257–1267.

Chen, L., Liu, C., Zou, R., Yang, M., Zhang, Z., 2016. Experimental examination of effectiveness of vegetation as bio-filter of particulate matters in the urban environment. Environ. Pollut. 208, 198–208.

Chen, L., Liu, C., Zhang, L., Zou, R., Zhang, Z., 2017. Variation in tree species ability to capture and retain airborne fine particulate matter (PM2.5). Sci. Rep. 7, 1–11.

Chong, K.Y., Tan, H.T.W., Corlett, R.T., 2009. A Checklist of the Total Vascular Plant Flora of Singapore Native, Naturalised and Cultivated Species. Raffles Museum of Biodiversity Research, National University of Singapore, Singapore, (273 pp).

Collins, T.J., 2007. ImageJ for microscopy. Biotechniques

Dzierzanowski, K., Gawronski, S.W., 2011. Use of trees for reducing particulate matter pollution in air. Challenges Mod. Technol. 2, 69–73.

Dzierżanowski, K., Popek, R., Gawrońska, H., Sæbø, A., Gawroński, S.W., 2011. Deposition of particulate matter of different size fractions on leaf surfaces and in waxes of urban forest species. Int. J. Phytoremediation 13, 1037–1046.

Fox, J., Weisberg, S., 2011. An {R} Companion to Applied Regression, second edition Sage, 2016.

Grote, R., Samson, R., Alonso, R., Amorim, J.H., Cariñanos, P., Churkina, G., Fares, S., Thiec, D. Le, Niinemets, , Mikkelsen, T.N., Paoletti, E., Tiwary, A., Calfapietra, C., 2016. Functional traits of urban trees: air pollution mitigation potential. Front. Ecol. Environ. 14, 543–550.

Hirabayashi, S., Kroll, C.N., Nowak, D.J., 2015. i-Tree Eco Dry Deposition Model Descriptions. 35.

Hofman, J., Bartholomeus, H., Calders, K., Van Wittenberghe, S., Wuyts, K., Samson, R., 2014. On the relation between tree crown morphology and particulate matter deposition on urban tree leaves: a ground-based LiDAR approach. Atmos. Environ. 99, 130–139.

Hwang, H.J., Yook, S.J., Ahn, K.H., 2011. Experimental investigation of submicron and ultrafine soot particle removal by tree leaves. Atmos. Environ. 45, 6987–6994.

i-Tree Eco. i-Tree Software Suite v6.x. (n.d.). http://www.itreetools.org/ (accessed 4 June 2019).

Koneig, J.Q., 2000. Health Effects of Ambient Air Pollution: How Safe Is the Air We Breathe?. Kluwer Academic Publishers, USA.

Li, W., Hopke, P.K., 1993. Initial size distributions and hygroscopicity of indoor combustion aerosol particles. Aerosol Sci. Technol. 19, 305–316.

Lin, M.-Y., Khlystov, A., 2011. Investigation of ultrafine particle deposition to vegetation branches in a wind tunnel. Aerosol Sci. Technol. 46, 465–472.

Lin, J.J., Noll, K.E., Holsen, T.M., 1994. Dry deposition velocities as a function of particle size in the ambient atmosphere. Aerosol Sci. Technol. 20, 239–252.

Lin, H.-Y., Hsu, D.-J., Su, J.-S., 2015. Particle size distribution of aromatic incense burning products. Int. J. Environ. Sci. Dev. 6, 857–860.

Litschike, T., Kuttler, W., 2008. On the reduction of urban particle concentration by vegetation - a review. Meteorol. Zeitschrift 17, 229–240.

Liu, L., Guan, D., Peart, M.R., 2012. The morphological structure of leaves and the dust-retaining capability of afforested plants in urban Guangzhou, South China. Environ. Sci. Pollut. Res. 19, 3440–3449.

Lung, S.C.C., Hu, S.C., 2003. Generation rates and emission factors of particulate matter and particle-bound polycyclic aromatic hydrocarbons of incense sticks. Chemosphere 50, 673–679.

Marcazzan, G.M., Vaccaro, S., Valli, G., Vecchi, R., 2001. Characterisation of PM10 and PM2.5 particulate matter in the ambient air of Milan (Italy). Atmos. Environ. 35, 4639–4650.

Mo, L., Ma, Z., Xu, Y., Sun, F., Lun, X., Liu, X., Chen, J., Yu, X., 2015. Assessing the capacity of plant species to accumulate particulate matter in Beijing, China. PLoS One 10, 1–18.

Mori, J., Sæbø, A., Hanslin, H.M., Teani, A., Ferrini, F., Fini, A., Burchi, G., 2015. Deposition of traffic-related air pollutants on leaves of six evergreen shrub species during a Mediterranean summer season. Urban For. Urban Green. 14, 264–273.

UNCORRECTED PROOF

Nowak, D.J., Hirabayashi, S., Bodine, A., Greenfield, E., 2014. Tree and forest effects on air quality and human health in the United States. Environ. Pollut. 193, 119–129.

Pant, P., Harrison, R.M., 2013. Estimation of the contribution of road traffic emissions to particulate matter concentrations from field measurements: a review. Atmos. Environ. 77, 78–97.

Pérez-Harguindeguy, N., Diaz, S., Garnier, E., Lavorel, S., Poorter, H., Jaureguiberry, P., Bret-Harte, M.S.S., Cornwell, W.K.K., Craine, J.M.M., Gurvich, D.E.E., Urcelay, C., Veneklaas, E.J.J., Reich, P.B.B., Poorter, L., Wright, I.J.J., Etc., Ray, P., Etc., Díaz, S., Lavorel, S., Poorter, H., Jaureguiberry, P., Bret-Harte, M.S.S., Cornwell, W.K.K., Craine, J.M.M., Gurvich, D.E.E., Urcelay, C., Veneklaas, E.J.J., Reich, P.B.B., Poorter, L., Wright, I.J.J., Ray, P., Enrico, L., Pausas, J.G., Vos, A.C. de, Buchmann, N., Funes, G., Quétier, F., Hodgson, J.G., Thompson, K., Morgan, H.D., Steege, H. ter, Heijden, M.G.A. van der, Sack, L., Blonder, B., Poschlod, P., Vaieretti, M. V., Conti, G., Staver, A.C., Aquino, S., Cornelissen, J.H.C., 2013. New handbook for standardized measurement of plant functional traits worldwide. Aust. J. Bot. 61, 167–234.

Popek, R., Gawronska, H., Gawronski, S.W., 2015. The level of particulate matter on foliage depends on the distance from the source of emission. Int. J. Phytoremediation 17, 1262–1268.

Prusty, B.A.K., Mishra, P.C., Azeez, P.A., 2005. Dust accumulation and leaf pigment content in vegetation near the national highway at Sambalpur, Orissa, India. Ecotoxicol. Environ. Saf. 60, 228–235.

Przybysz, A., Sæbø, A., Hanslin, H.M., Gawroński, S.W., 2014. Accumulation of particulate matter and trace elements on vegetation as affected by pollution level, rainfall and the passage of time. Sci. Total Environ. 381, 360–369.

Quah, E., Boon, T.L., 2003. The economic cost of particulate air pollution on health in Singapore. J. Asian Econ. 14, 73–90.

R Core Team, 2017. R Development Core Team. R A Lang. Environ. Stat. Comput http://www.R-project.org.

Räsänen, J.V., Yli-Pirilä, P., Holopainen, T., Joutsensaari, J., Pasanen, P., Kivimäenpää, M., 2012. Soil drought increases atmospheric fine particle capture efficiency of Norway spruce. Boreal Environ. Res. 17, 21–30.

Räsänen, J.V., Holopainen, T., Joutsensaari, J., Ndam, C., Pasanen, P., Rinnan, , Kivimäenpää, M., 2013. Effects of species-specific leaf characteristics and reduced water availability on fine particle capture efficiency of trees. Environ. Pollut. 183, 64–70.

Reich, B., Uhl, C., Waiters, B., Ellsworth, S., 1991. Leaf lifespan as a determinant of leaf structure and function among 23 amazonian tree species. Oecologia 16–24.

Sæbø, A., Popek, R., Nawrot, B., Hanslin, H.M., Gawronska, H., Gawronski, S.W., 2012. Plant species differences in particulate matter accumulation on leaf surfaces. Sci. Total Environ. 427–428, 347–354.

Selmi, W., Weber, C., Rivière, E., Blond, N., Mehdi, L., Nowak, D., 2016. Air pollution removal by trees in public green spaces in Strasbourg city, France. Urban For. Urban Green. 17, 192–201.

Siao, W.S., Balasubramanian, R., Rianawati, E., Karthikeyan, S., Streets, D.G., 2007. Characterization and source apportionment of particulate matter $\leq 2.5\,\mu m$ in Sumatra, Indonesia, during a recent peat fire episode. Environ. Sci. Technol. 41, 3488–3494.

Singapore Department of Statistics, 2017. Yearbook of Statistics Singapore 2017. Department of Statistics, Ministry of Trade & Industry, Republic of Singapore, Singapore.

Singapore Department of Statistics, 2018. Yearbook of Statistics Singapore, 2018. Department of Statistics, Ministry of Trade & Industry, Republic of Singapore, Singapore.

Slinn, W.G.N., 1982. Predictions for particle deposition to vegetative canopies. Atmos. Environ. 16, 1785–1794.

Song, Y., Maher, B.A., Li, F., Wang, X., Sun, X., Zhang, H., 2015. Particulate matter deposited on leaf of five evergreen species in Beijing, China: source identification and size distribution. Atmos. Environ. 105, 53–60.

Song, X.P., Richards, D., Edwards, P., Tan, P.Y., 2017. Benefits of trees in tropical cities. Science 356, 1241.

Song, X.P., Tan, P.Y., Edwards, P., Richards, D., 2018. The economic benefits and costs of trees in urban forest stewardship: a systematic review. Urban For. Urban Green. 29, 162–170.

Tallis, M., Taylor, G., Sinnett, D., Freer-Smith, P., 2011. Estimating the removal of atmospheric particulate pollution by the urban tree canopy of London, under current and future environments. Landsc. Urban Plan. 103, 129–138.

U.S. Department of Agriculture Forest Service, i-Tree Eco, International. In: www.itreetools.org/eco/international.phpAccessed 30 May 2019.

Vu, T.V., Ondracek, J., Zdímal, V., Schwarz, J., Delgado-Saborit, J.M., Harrison, R.M., 2017. Physical properties and lung deposition of particles emitted from five major indoor sources. Air Qual. Atmos. Heal. 10, 1–14.

Wang, L., Liu, L.Y., Gao, S.Y., Hasi, E., Wang, Z., 2006. Physicochemical characteristics of ambient particles settling upon leaf surfaces of urban plants in Beijing. J. Environ. Sci. (China) 18, 921–926.

Weber, F., Kowarik, I., Saumel, I., 2014. Herbaceous plants as filters: immobilization of particulates along urban street corridors. Environ. Pollut. 186, 234–240.

Weerakkody, U., Dover, J.W., Mitchell, P., Reiling, K., 2018. Evaluating the impact of individual leaf traits on atmospheric particulate matter accumulation using natural and synthetic leaves. Urban For. Urban Green. 30, 98–107.

WHO, 2002. The World Health Report 2002. 16, World Heal. Organ, 244.

WHO, 2006. WHO Air Quality Guidelines for Particulate Matter, Ozone, Nitrogen Dioxide and Sulfur Dioxide: Global Update 2005: Summary of Risk Assessment. Geneva World Heal. Organ, 1–22.

WHO, 2016. Ambient Air Pollution: A Global Assessment of Exposure and Burden of Disease. World Heal. Organ.

WHO, 2017. Evolution of WHO Air Quality Guidelines.

Wong, C.-M., Vichit-Vadakan, N., Kan, H., Qian, Z., 2008. Public health and Air Pollution in Asia (PAPA): a multicity study of short-term effects of air pollution on mortality. Environ. Health Perspect. 116, 1195–1202.

Yang, J., McBride, J., Zhou, J., Sun, Z., 2004. The urban forest in Beijing and its role in air pollution reduction. Urban For. Urban Green. 3, 65–78.

Yang, C.R., Lin, T.C., Chang, F.H., 2007. Particle size distribution and PAH concentrations of incense smoke in a combustion chamber. Environ. Pollut. 145, 606–615.

Yli-Pelkonen, V., Setälä, H., Viippola, V., 2017. Urban forests near roads do not reduce gaseous air pollutant concentrations but have an impact on particles levels. Landsc. Urban Plan. 158, 39–47.

Yuen, J.Q., Olin, P.H., Lim, H.S., Benner, S.G., Sutherland, R.A., Ziegler, A.D., 2012. Accumulation of potentially toxic elements in road deposited sediments in residential and light industrial neighborhoods of Singapore. J. Environ. Manag. 101, 151–163.

UNCORRECTED PROOF